

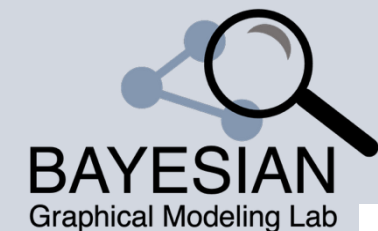
Bayesian Graphical Modeling

Model Uncertainty in Psychometrics

What is psychometrics? 2026

m.marsman@uva.nl

bayesiangraphicalmodeling.com



Bayesian Graphical Modeling

https://github.com/Bayesian-Graphical-Modelling-Lab/BGM_Workshops/tree/WiP_2026

What is psychometrics? 2026

m.marsman@uva.nl

bayesiangraphicalmodeling.com



Session setup



Part I: Theory

Session setup



Part I: Theory



Part II: Tutorial

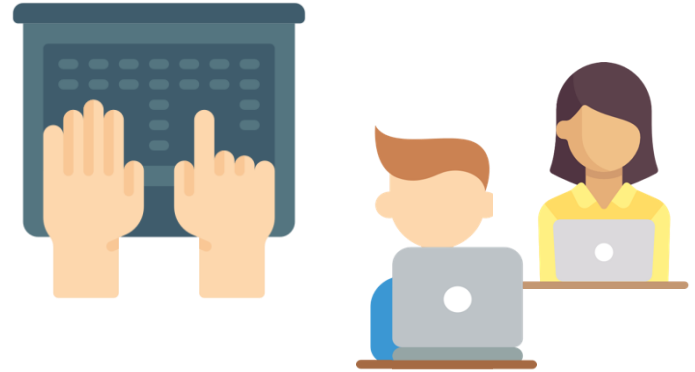
Session setup



Part I: Theory



Part II: Tutorial



Part III: Tutorial

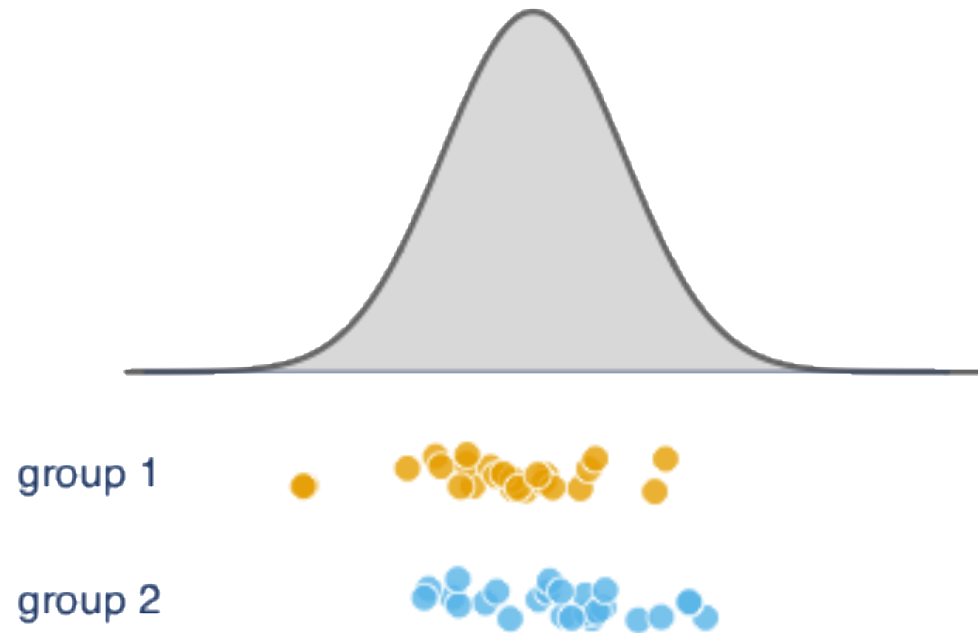
Session setup



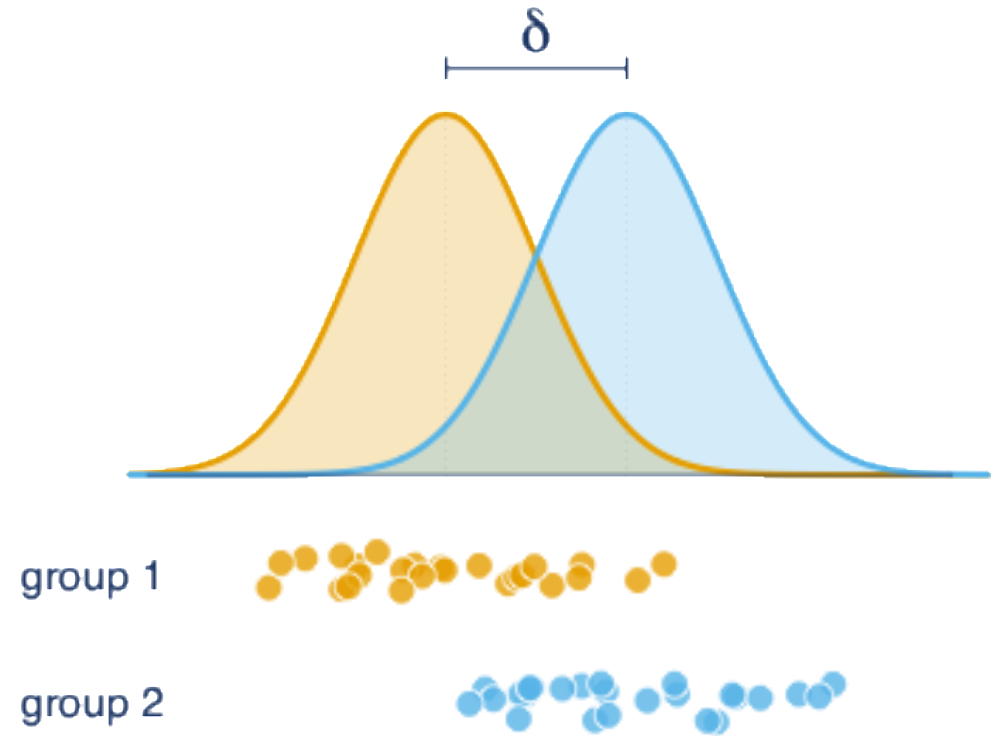
Part I: Theory

Model space: T-test

$$\mathcal{M}_0: \delta = 0$$



$$\mathcal{M}_1: \delta \neq 0$$



Model space: ANOVA

$$\mathcal{M}_0 : \mu_{jk} = \mu$$

$$\mathcal{M}_1 : \mu_{jk} = \mu + \alpha_j$$

$$\mathcal{M}_2 : \mu_{jk} = \mu + \beta_k$$

$$\mathcal{M}_3 : \mu_{jk} = \mu + \alpha_j + \beta_k$$

$$\mathcal{M}_4 : \mu_{jk} = \mu + \alpha_j + \beta_k + (\alpha\beta)_{jk}$$

Model space: ANOVA

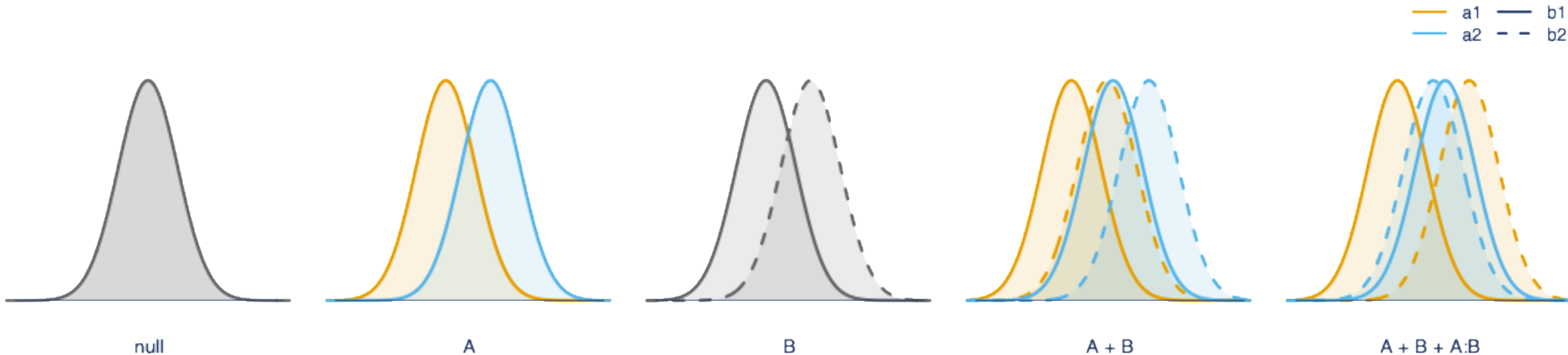
$$\mathcal{M}_0 : \mu_{jk} = \mu$$

$$\mathcal{M}_1 : \mu_{jk} = \mu + \alpha_j$$

$$\mathcal{M}_2 : \mu_{jk} = \mu + \beta_k$$

$$\mathcal{M}_3 : \mu_{jk} = \mu + \alpha_j + \beta_k$$

$$\mathcal{M}_4 : \mu_{jk} = \mu + \alpha_j + \beta_k + (\alpha\beta)_{jk}$$



Model space: ANOVA

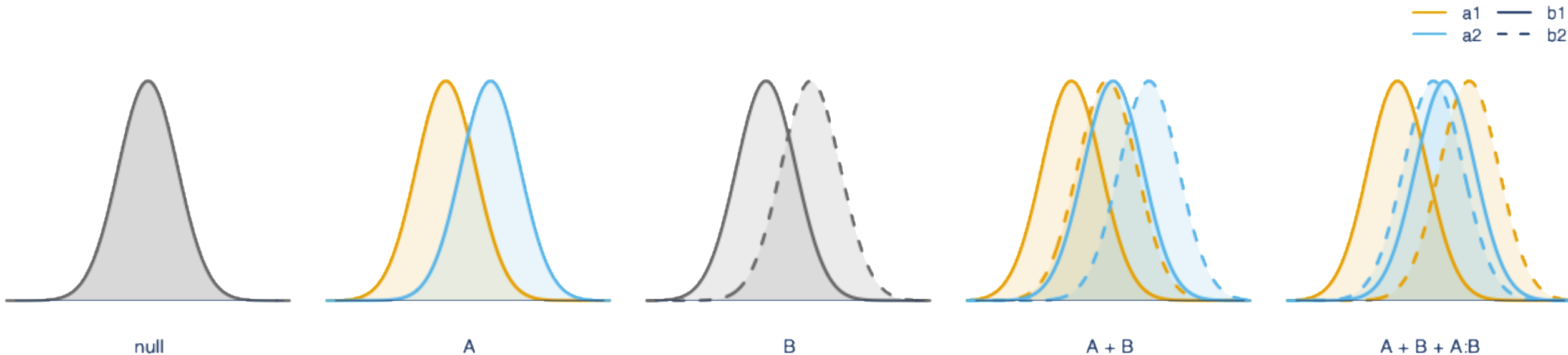
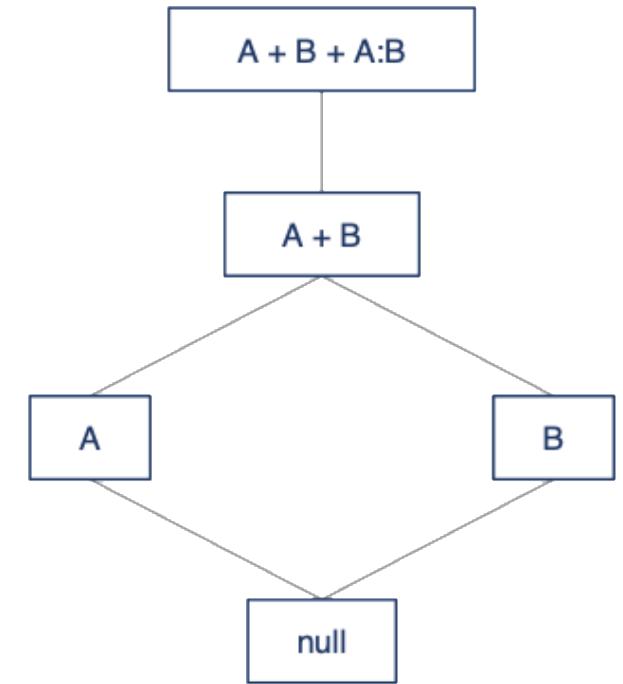
$$\mathcal{M}_0 : \mu_{jk} = \mu$$

$$\mathcal{M}_1 : \mu_{jk} = \mu + \alpha_j$$

$$\mathcal{M}_2 : \mu_{jk} = \mu + \beta_k$$

$$\mathcal{M}_3 : \mu_{jk} = \mu + \alpha_j + \beta_k$$

$$\mathcal{M}_4 : \mu_{jk} = \mu + \alpha_j + \beta_k + (\alpha\beta)_{jk}$$



Model space: Regression

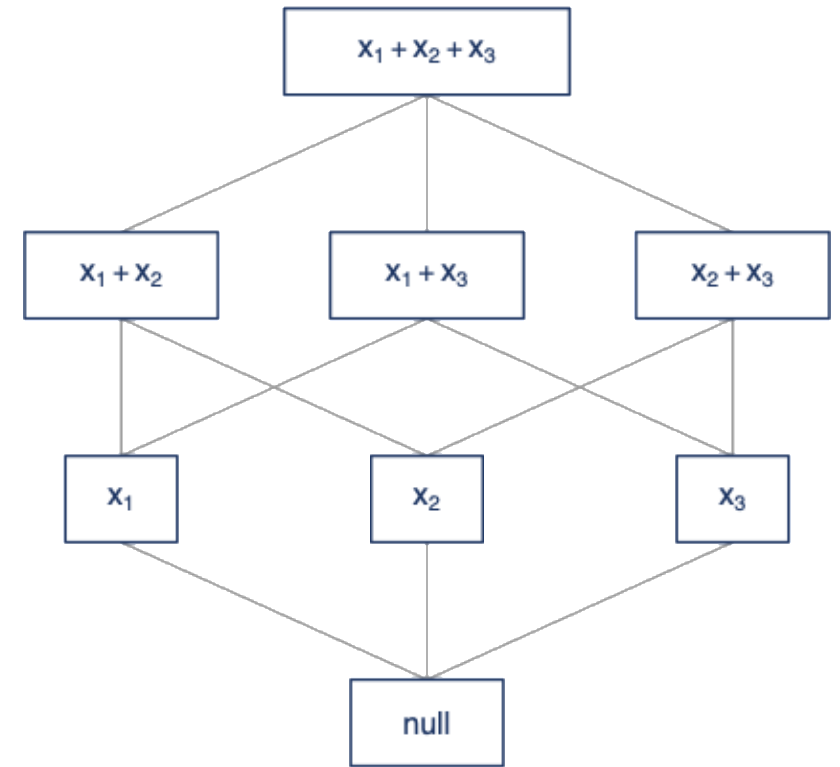
$$y_i \sim N \left(\beta_0 + \sum_{j=1}^p \gamma_j \beta_j x_{ij}, \sigma^2 \right), \gamma_j \in \{0, 1\}$$

Model space: Regression

$$y_i \sim N \left(\beta_0 + \sum_{j=1}^p \gamma_j \beta_j x_{ij}, \sigma^2 \right), \gamma_j \in \{0, 1\}$$

$\mathcal{M}_\gamma, \gamma = (\gamma_1, \dots, \gamma_p) \in \{0, 1\}^p, 2^p$ models

Model space: Regression



$$y_i \sim N \left(\beta_0 + \sum_{j=1}^p \gamma_j \beta_j x_{ij}, \sigma^2 \right), \gamma_j \in \{0, 1\}$$

$\mathcal{M}_\gamma, \gamma = (\gamma_1, \dots, \gamma_p) \in \{0, 1\}^p, 2^p$ models

What is your model space?

Discuss with your neighbor:

1. Formulate the model or design you are working with.
2. List the choices in it that could have been made differently.
3. How many possible models does that give?

3 minutes, then we discuss as a group

Example model spaces in psychometrics

Structural equation models: Paths, loadings, modification indices,

Item response theory: Discrimination, guessing, item-factor loadings, differential item functioning

Longitudinal models: Lags, cross-lagged paths, random effects, covariates

Multiverse analysis: Every analysis decision (e.g., outlier removal, variable transformation, covariates, cutoffs, etc.)

My model space

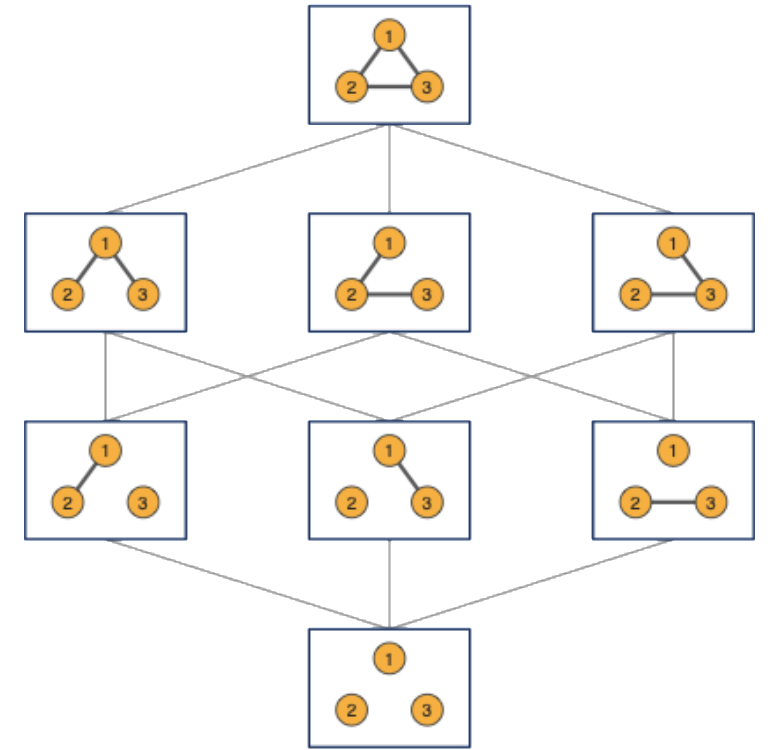
$$p(\mathbf{x}) \propto \exp \left(\sum_i \mu_i(x_i) + \sum_{i < j} \gamma_{ij} \sigma_{ij} x_i x_j \right), \gamma_{ij} \in \{0, 1\}$$

My model space

$$p(\mathbf{x}) \propto \exp \left(\sum_i \mu_i(x_i) + \sum_{i < j} \gamma_{ij} \sigma_{ij} x_i x_j \right), \gamma_{ij} \in \{0, 1\}$$

p nodes, $\binom{p}{2} = \frac{p(p-1)}{2}$ candidate edges, $2^{p(p-1)/2}$ structures

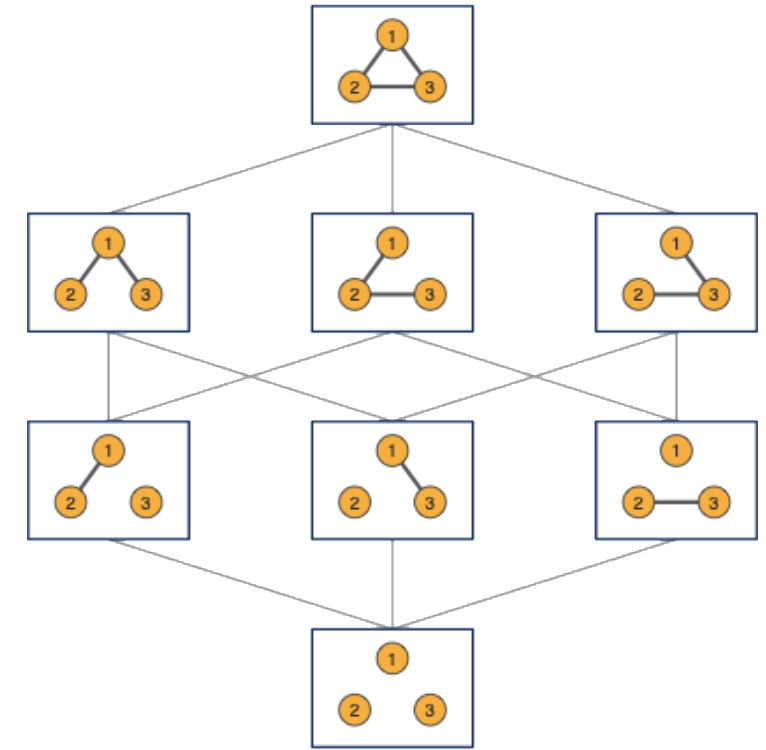
My model space



$$p(\mathbf{x}) \propto \exp \left(\sum_i \mu_i(x_i) + \sum_{i < j} \gamma_{ij} \sigma_{ij} x_i x_j \right), \gamma_{ij} \in \{0, 1\}$$

p nodes, $\binom{p}{2} = \frac{p(p-1)}{2}$ candidate edges, $2^{p(p-1)/2}$ structures

My model space



p	Regression, 2^p	Cross-sectional, $2^{p \times (p-1) / 2}$	Longitudinal, $2^{p \times p}$
3	8	8	512
9	512	68,719,476,736	2,417,851,639,229,258,349,412,352
17	131,072	87,112,285,931,760,246,646,623,89 9,502,532,662,132,736	994,646,472,819,573,284,310,764,496,293,64 1,680,200,912,301,594,695,434,880,927,953,7 86,318,994,025,066,751,066,112

Model or effect selection

$$\underbrace{\frac{p(\mathcal{M}_1 \mid \text{data})}{p(\mathcal{M}_0 \mid \text{data})}}_{\text{Posterior Odds}} = \text{BF}_{10} \times \underbrace{\frac{p(\mathcal{M}_1)}{p(\mathcal{M}_0)}}_{\text{Prior Odds}}$$

Model or effect selection

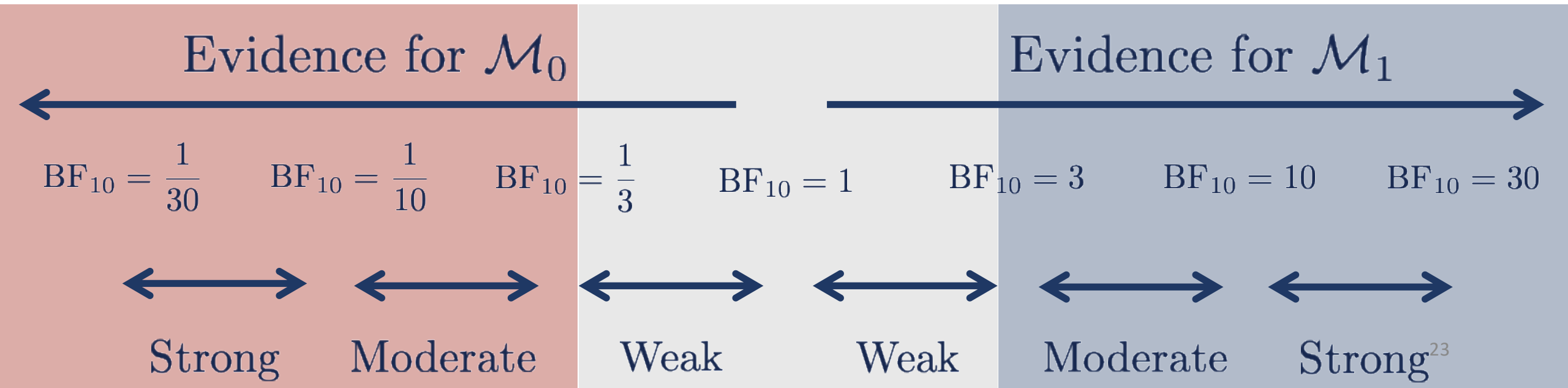
$$\underbrace{\frac{p(\mathcal{M}_1 \mid \text{data})}{p(\mathcal{M}_0 \mid \text{data})}}_{\text{Posterior Odds}} = \overbrace{\text{BF}_{10}}^{\text{Bayes factor}} \times \underbrace{\frac{p(\mathcal{M}_1)}{p(\mathcal{M}_0)}}_{\text{Prior Odds}}$$

Model or effect selection

$$\text{BF}_{10} = \frac{p(\text{data} \mid \mathcal{M}_1)}{p(\text{data} \mid \mathcal{M}_0)}$$

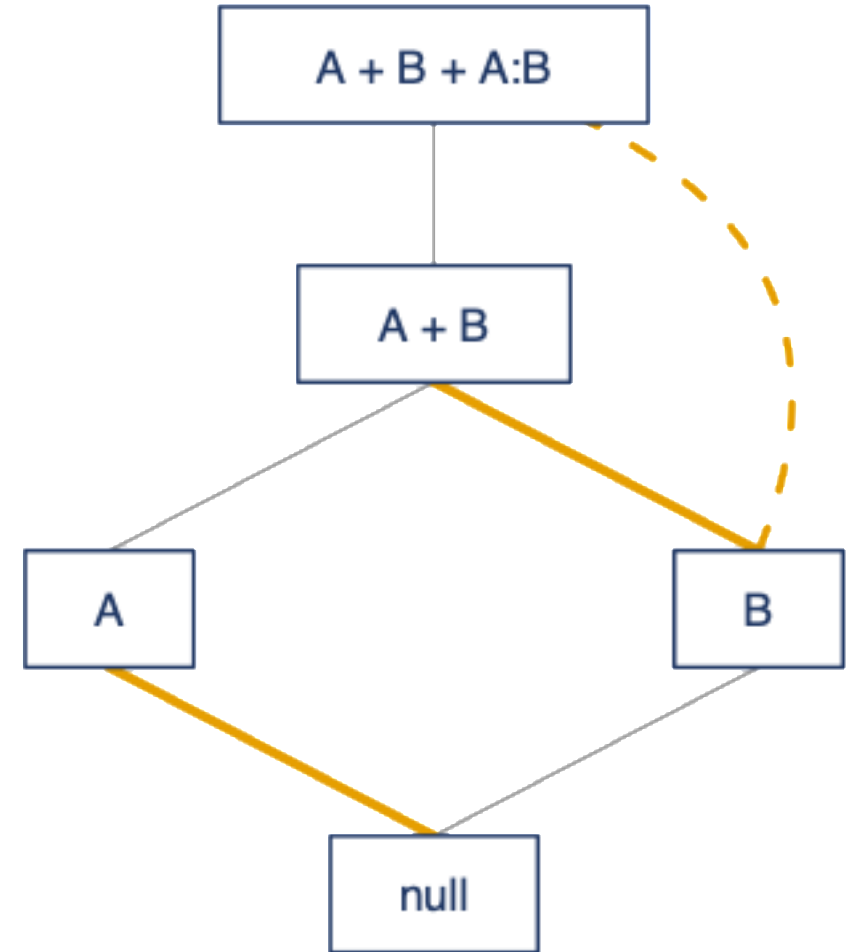
Model or effect selection

$$\text{BF}_{10} = \frac{p(\text{data} \mid \mathcal{M}_1)}{p(\text{data} \mid \mathcal{M}_0)}$$



Model or effect selection: ANOVA

$\text{BF}_{\mathcal{M}_1:\mathcal{M}_0}$ and $\text{BF}_{\mathcal{M}_3:\mathcal{M}_2}$



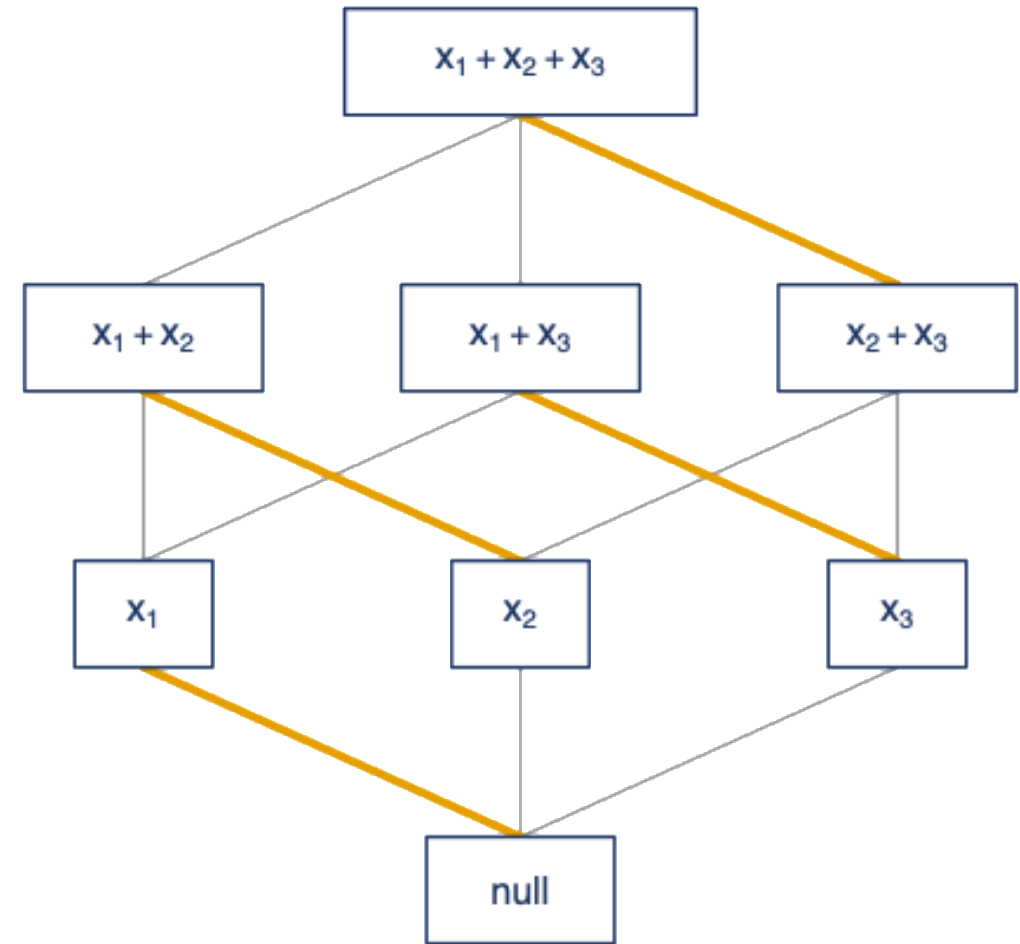
Model or effect selection: Regression

$\text{BF}_{\mathcal{M}_{100} : \mathcal{M}_{000}},$

$\text{BF}_{\mathcal{M}_{110} : \mathcal{M}_{010}},$

$\text{BF}_{\mathcal{M}_{101} : \mathcal{M}_{001}},$

$\text{BF}_{\mathcal{M}_{111} : \mathcal{M}_{011}}$

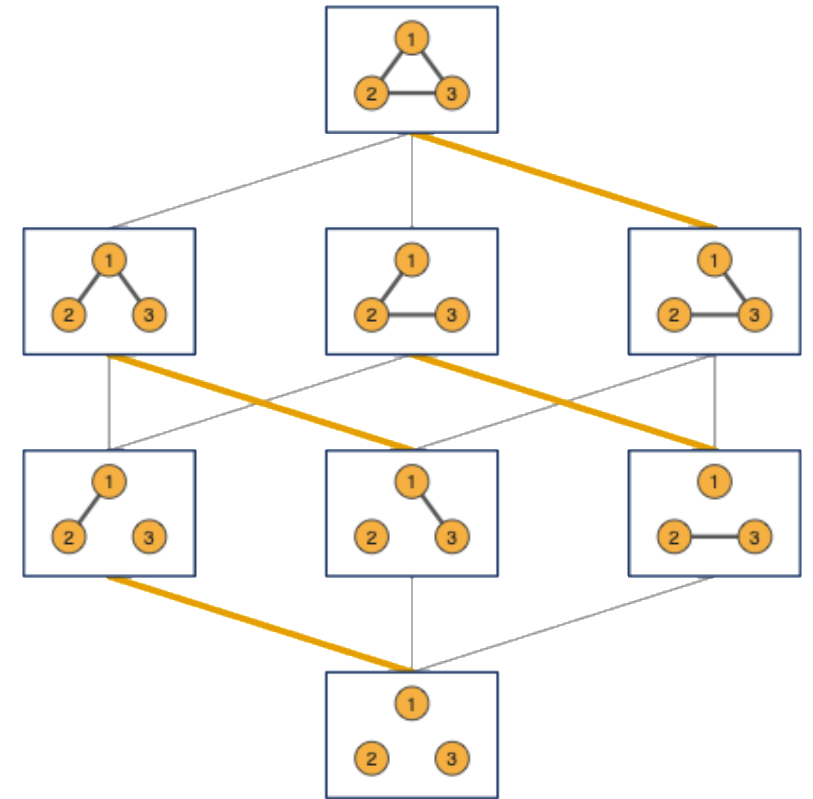


Model or effect selection: Graphical model

Single model Bayes factor:

$$\text{BF}_{\mathcal{M}_{111} : \mathcal{M}_{011}}$$

Test exclusion against all other edges included.
(One of four possible comparisons.)

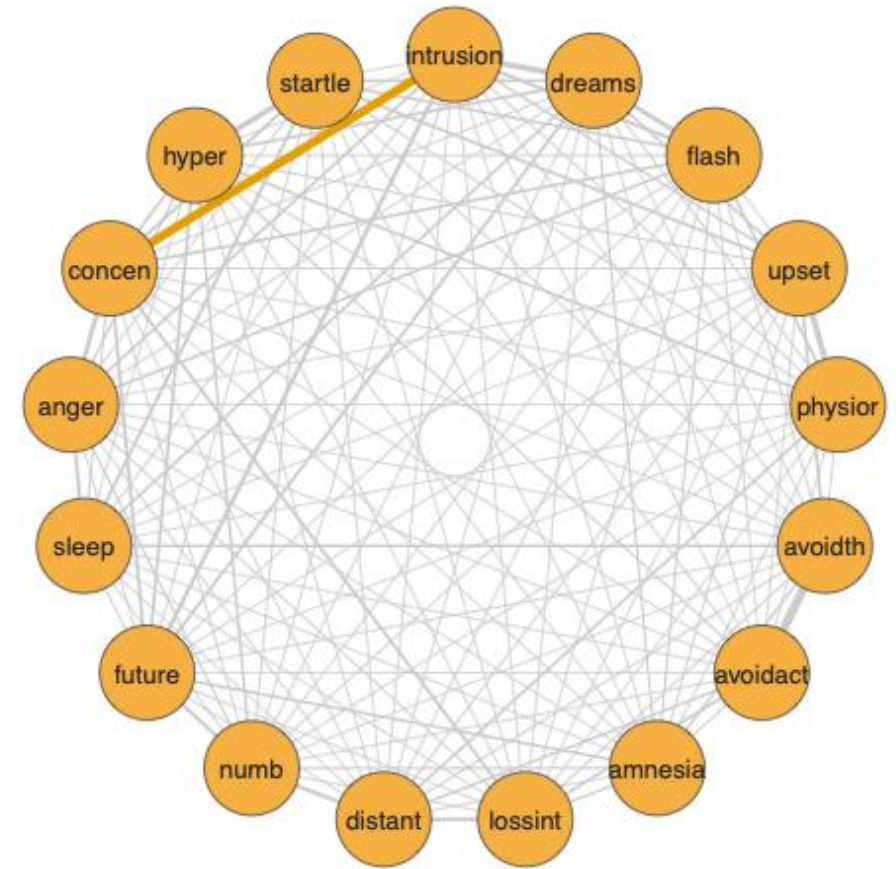


Model or effect selection: Graphical model

Single model Bayes factor:

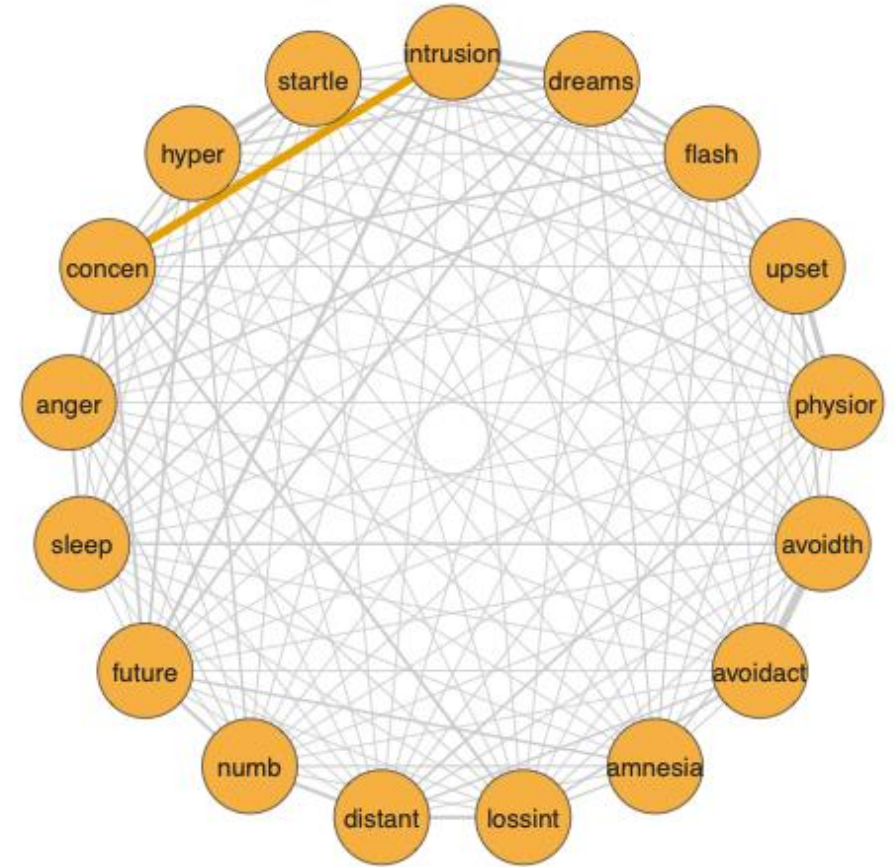
$$\text{BF} = \frac{p(\text{data} \mid \gamma_{ij} = 1, \gamma_{-ij} = 1)}{p(\text{data} \mid \gamma_{ij} = 0, \gamma_{-ij} = 1)}$$

Test exclusion against all other edges included.
(One of 2^{135} possible comparisons.)



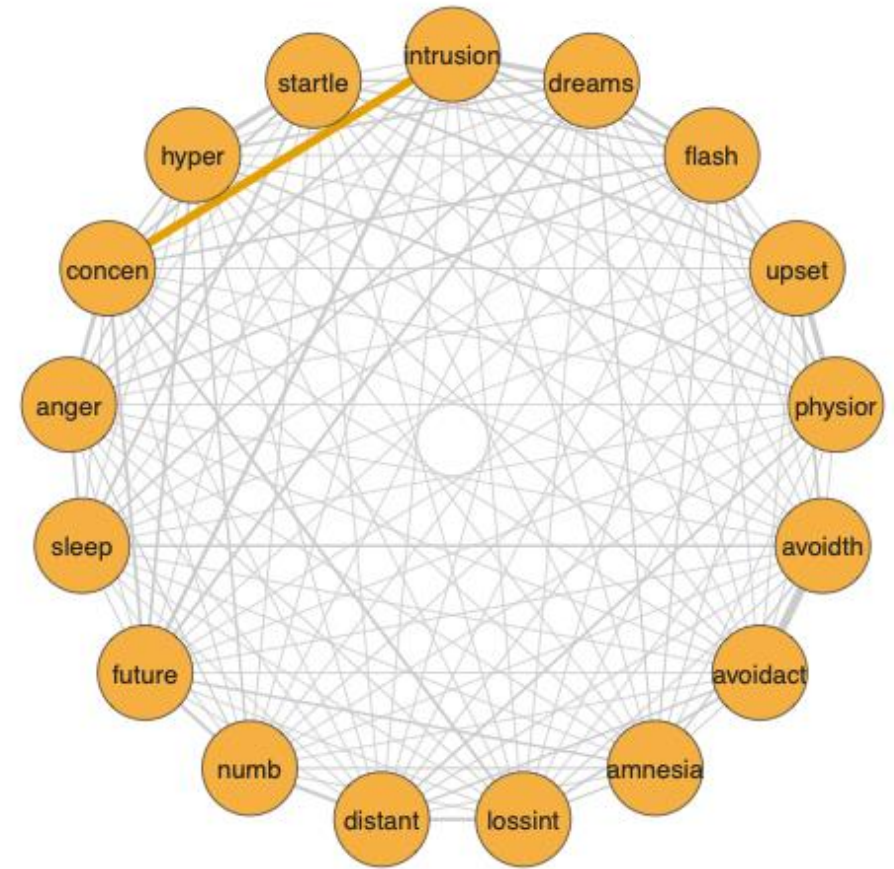
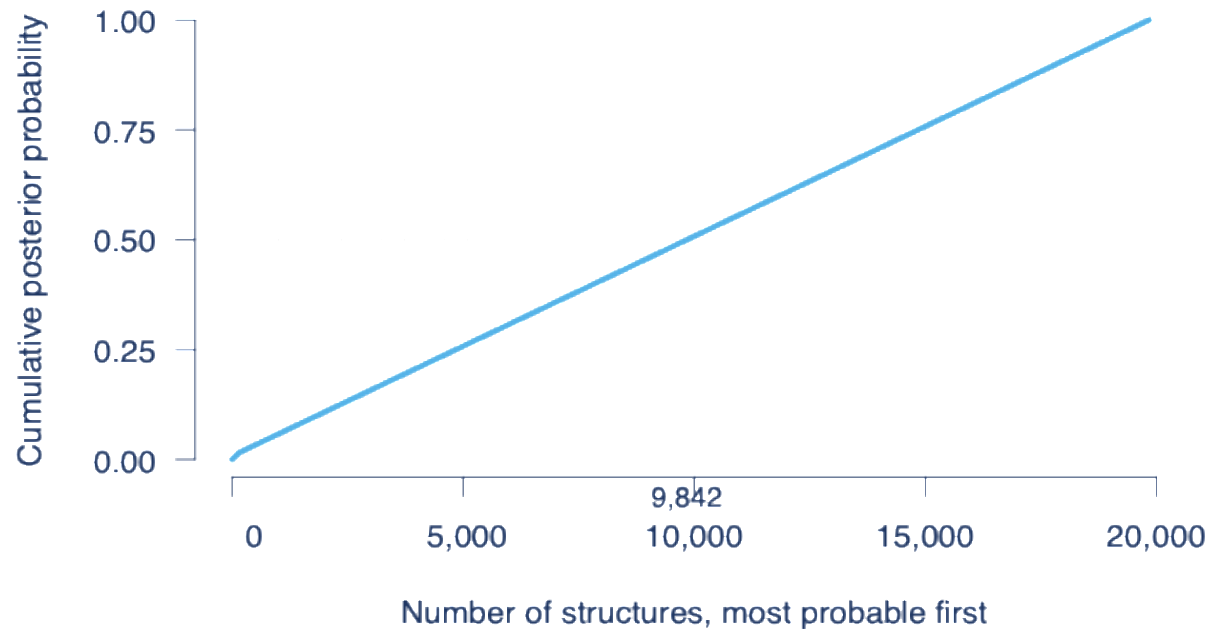
What to report?

$$BF = \frac{p(\text{data} \mid \gamma_{ij} = 1, \gamma_{-ij} = \mathbf{1})}{p(\text{data} \mid \gamma_{ij} = 0, \gamma_{-ij} = \mathbf{1})}$$



Model uncertainty

Different models are similarly plausible
No clear best model



Ignoring model uncertainty

Selected model is treated as known

- Inference is conditional on this selection:
 - Confidence or credible intervals are too narrow
 - Overconfidence in effects and their size

Conclusions depend on which model is selected

- Different sample, different model selected

Model averaging: Posterior inclusion probability

$\mathcal{H}_0$: there is no edge between variables i and j ($\gamma_{ij} = 0$)

$\mathcal{H}_1$: there is an edge between variables i and j ($\gamma_{ij} = 1$)

$$p(\mathcal{H}_1 \mid \text{data}) = p(\gamma_{ij} = 1 \mid \text{data}) = \sum_{\boldsymbol{\gamma}: \gamma_{ij}=1} p(\boldsymbol{\gamma} \mid \text{data})$$

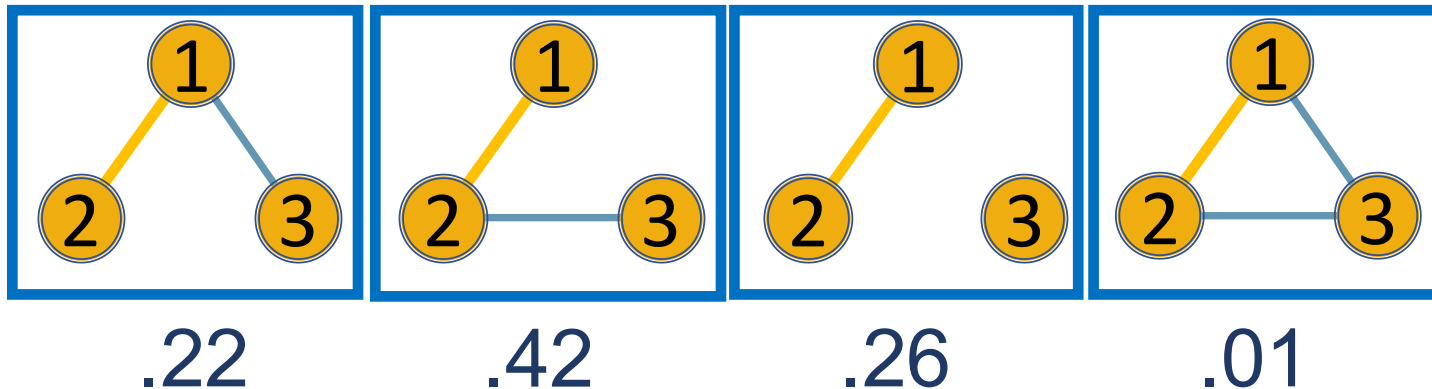
Model averaging: Posterior inclusion probability

$\mathcal{H}_0$: there is no edge between variables i and j ($\gamma_{ij} = 0$)

$\mathcal{H}_1$: there is an edge between variables i and j ($\gamma_{ij} = 1$)

$$p(\mathcal{H}_0 \mid \text{data}) = 1 - p(\mathcal{H}_1 \mid \text{data}) = 0.09$$

$$p(\mathcal{H}_1 \mid \text{data}) = 0.91$$



Model averaging: Inclusion Bayes factor

$\mathcal{H}_0$: there is no edge between variables i and j ($\gamma_{ij} = 0$)

$\mathcal{H}_1$: there is an edge between variables i and j ($\gamma_{ij} = 1$)

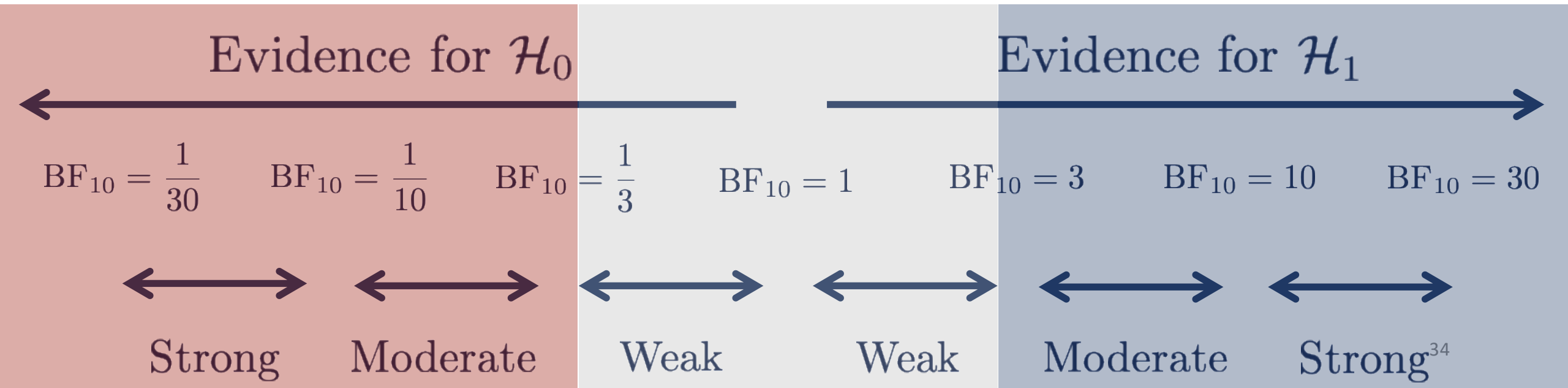
$$\text{BF}_{\text{incl}} = \frac{p(\mathcal{H}_1 \mid \text{data}) / p(\mathcal{H}_0 \mid \text{data})}{p(\mathcal{H}_1) / p(\mathcal{H}_0)}$$

$$\frac{P(\mathcal{H}_1 \mid \text{data})}{P(\mathcal{H}_0 \mid \text{data})} = \frac{.91}{.09} = 10.11$$

$$\frac{p(\mathcal{H}_1)}{p(\mathcal{H}_0)} = 1$$

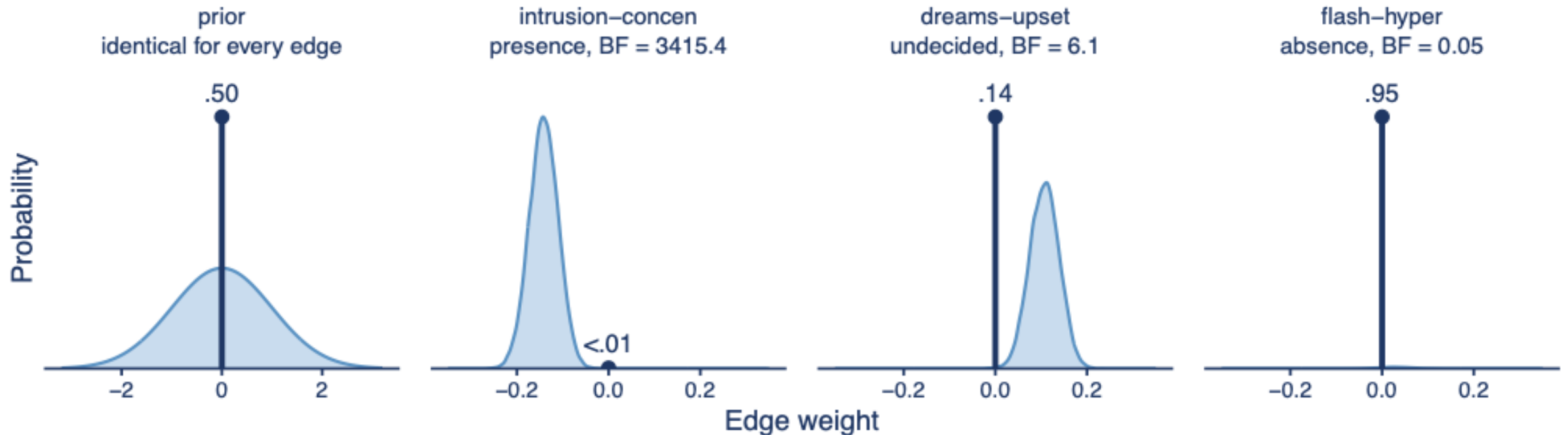
Model averaging: Inclusion Bayes factor

$$\text{BF}_{10} = \frac{P(\mathcal{H}_1 \mid \text{data})}{P(\mathcal{H}_0 \mid \text{data})} \bigg/ \frac{P(\mathcal{H}_1)}{P(\mathcal{H}_0)} = 10.11$$



Model averaging: Posterior

$$p(\theta \mid \text{data}) = \sum_{\gamma'} p(\theta \mid \text{data}, \gamma') p(\gamma' \mid \text{data})$$



Model averaging

Benefits

- Model uncertainty carries into every estimate and test
- One test per effect, not per condition
- More robust predictions than under a single model

Costs

- Challenging to define good model priors
- Assumes correct model is in the model space
- Computationally challenging
- What to plot?

What is your model space?

Discuss with your neighbor (2 minutes)

1. What would a single-model comparison for your model look like?
2. Would model averaging be worthwhile for your model, and why?

Session setup



Part I: Theory

Session setup



Part I: Theory



Part II: Tutorial